

30
DAYS

Generative AI

Learning Roadmap

Python ✓

SQL ✓

2hrs/day

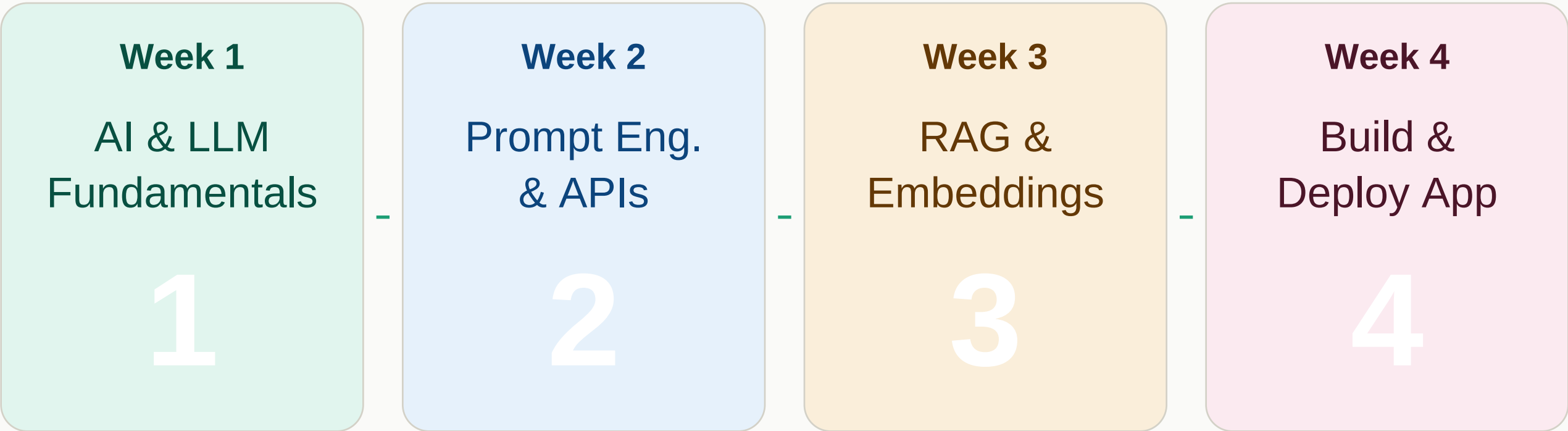
For Python + SQL professionals

Swipe to explore →



30-Day GenAI Roadmap

Your learning journey



28 daily tasks **4** projects **8+** resources **1** live app

Week 1

AI & LLM Fundamentals

Day 1

What is Generative AI?

Watch Andrej Karpathy's Intro to LLMs. Write 5 key takeaways.

■ Video

Day 2

How transformers work

Watch 3Blue1Brown's Attention video. Sketch the architecture.

■ Video

Day 3

Set up your Python env

Install openai, anthropic, python-dotenv. Run Hello World API call.

👨🍳 Practice

Day 4

First API call

Write a script calling GPT-4o or Claude. Tweak temperature settings.

👨🍳 Practice

Day 5

Tokens & context windows

Count tokens in your SQL queries using tiktoken library.

■ Video

Day 6

OpenAI Cookbook

Browse 3 examples. Adapt one to work with your own sample data.

■ Project

Day 7

Week 1 mini-project

Build a CLI chatbot that answers questions about a SQL table.

■ Project

Week 2

Prompt Engineering & APIs

Day 8

Prompt engineering basics

Watch DeepLearning.AI Prompt Engineering course — Lessons 1 & 2.

■ Video

Day 9

Zero-shot vs few-shot

Write 10 prompt variations. Measure which produces best SQL output.

📝 Practice

Day 10

Chain-of-thought prompting

Apply CoT to get LLMs to explain SQL query logic step by step.

■ Video

Day 11

System prompts & roles

Build an SQL expert assistant with a custom system prompt.

📝 Practice

Day 12

Structured outputs

Get LLM to return JSON you can load into a Pandas DataFrame.

📝 Practice

Day 13

LangChain intro

Watch LangChain crash course. Build a chain: question → SQL query.

■ Video

Day 14

Week 2 mini-project

Natural language → SQL converter using your existing SQL skills.

■ Project

Week 3

RAG & Embeddings

Day 15

What is RAG?

Watch IBM's RAG explainer. Read the original RAG paper abstract.

■ Video

Day 16

Embeddings explained

Compute embeddings for 20 sentences. Visualize similarity scores.

■ Video

Day 17

Vector databases

Set up ChromaDB locally. Store 100 chunks — like a semantic SQL WHERE.

👉 Practice

Day 18

Build your first RAG pipeline

Ingest a PDF, chunk it, embed it, store in Chroma. Query it.

■ Project

Day 19

RAG evaluation

Run 10 Q&A pairs through your pipeline. Score with RAGAS metrics.

👉 Practice

Day 20

Advanced RAG patterns

Implement hybrid search (keyword + semantic) on your project.

■ Video

Day 21

Week 3 mini-project

'Chat with your database' — NL→SQL + RAG for unstructured data.

■ Project

Week 4

Build & Ship a Real GenAI App

Day 22

Agents & tool use

Build an agent that queries a DB or searches the web as needed.

■ Video

Day 23

LLM evals & testing

Set up an eval harness with 20 test cases. Log pass/fail rates.

👉 Practice

Day 24

Streamlit UI

Wrap your RAG app in Streamlit. Upload a file, ask questions.

■ Project

Day 25

Prompt versioning

Use Langfuse to track prompt changes. Version your best prompts.

👉 Practice

Day 26

Cost & latency tuning

Profile your app. Compare GPT-4o-mini vs GPT-4o tradeoffs.

👉 Practice

Day 27

Deploy to HuggingFace

Deploy your Streamlit app to HuggingFace Spaces — free & live!

■ Project

Day 28

Polish & document

Write a README. Record a 2-min Loom demo. Update LinkedIn.

■ Project

4 Hands-On Projects

One per week — cumulative & portfolio-ready

01 CLI Chatbot

Week 1

Python API call → conversational Q&A over a CSV/SQL tab

02 NL → SQL Converter

Week 2

Type English, get runnable SQL. Uses your SQL expertise

03 Chat with Your DB

Week 3

RAG pipeline: embed docs + query with natural language.

04 Deployed App

Week 4

Streamlit UI + HuggingFace deploy. Live URL for your po

Essential Resources

All free to start

■ DeepLearning.AI short course

Best free GenAI video courses

■ Andrej Karpathy — YouTube

Deep LLM intuition, visually explained

■ OpenAI Cookbook (GitHub)

Production Python patterns & examples

■ LangChain docs + tutorials

RAG, agents, chains — excellent quickstart

■ fast.ai Practical Deep Learn

Code-first, project-first approach

■ Hugging Face NLP Course

Free certificate, hands-on notebooks

After 30 Days You'll Have

LLM API integration

Prompt engineering

RAG pipelines

Vector databases

LangChain

Streamlit UIs

LLM evaluation

Responsible AI

Agent patterns

Cost optimization

Hugging Face

Structured outputs

Portfolio: 4 real projects · 1 live deployed app

You're no longer a beginner.

You're a GenAI practitioner.